

DIGITAL PRODUCT DESIGN

Lecture 6: Platforms and Network Effects

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Outline

- Pricing Review
- Network Effects
- Platform Strategy – Launch
- Platform Strategy – Monetization

PRICING REVIEW

Pricing Strategies

- Cost-based pricing – Bad
- Competition-based pricing – Bad
- Customer-based pricing – Bad

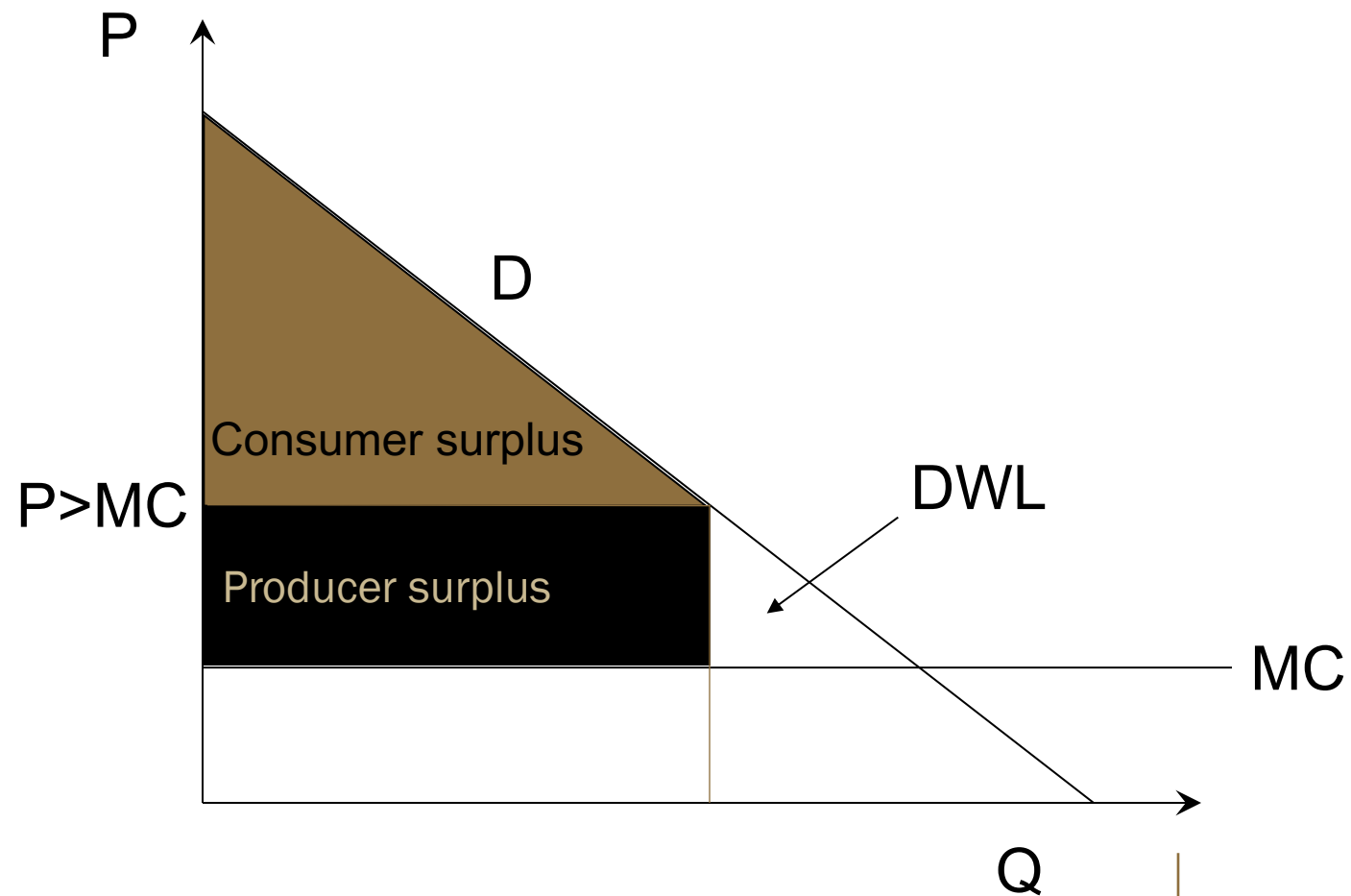
- Value based pricing – Good

Value Based Pricing

- Estimate a demand curve
 - Demand is the quantity sold at a specific price $Q(P)$
- Economic theory: People purchase if their value for product exceeds the price
 - Demand aggregates individual valuations: $Q(P) = \sum_i 1[V_i > P]$
- Set price to maximize profits:

$$P^* = \arg \max_P P * Q(P) - C(Q(P))$$

Pricing with a Demand Curve



Price Elasticity is a Key Quantity

- Price elasticity of demand:

$$E_d = \frac{\% \text{ Change in Quantity Demanded}}{\% \text{ Change in Price}} = \frac{\Delta Q / Q}{\Delta P / P}$$

- Under certain conditions:

$$\frac{P^* - MC}{P^*} = -\frac{1}{E_d}$$

- Implications

- If elasticity is close to 0, optimal markups can be large
- If people are price elastic (E_d very negative) optimal markups smaller

Using Data to Segment Customers

- We can also try to predict our high-value customers
 - Customers who value product a bunch
 - Also known as price-inelastic customers
 - If we raise prices, they still purchase
- If we can identify these customers, potential to charge them a different price
 - Known as personalized pricing or “price discrimination”
 - Common methods: Student discount, changing prices over time

Bundling

- Bundling allows you to (potentially) extract value in a world with heterogeneous preferences
- Idea is value for the bundle is often more stable than value of individual goods
 - Can be formalized using a law of large numbers argument
- This allows bundling strategies to extract more value
 - Often relies on zero marginal costs and free disposal

MOTIVATION FOR STUDYING PLATFORMS

Platforms

- “A platform is a business based on enabling value-creating interactions between external producers and consumers.”
- Parker, Van Alstyne & Choudary
- Platforms provide infrastructure & governance
 - Anyone can participate within certain boundaries
 - Platform facilitates matches among participants

Platform Companies Are Ubiquitous in Economy

Industry	Examples
Communication & Networking	LinkedIn, Facebook, Twitter, Tinder, ...
Consumer Goods	Philips, McCormick Foods FlavorPrint
Education	Udemy, Skillshare, Coursera
Energy	Nest
Finance	Bitcoin, Kickstarter
Health Care	Cohealo, SimplyInsured, Kaiser Permanente
Gaming	Xbox, Nintendo, PlayStation
Labor	Upwork, Fiverr, ...
Local Services	Yelp, Foursquare, Groupon
Media	Medium, YouTube, Wikipedia
OS	iOS, Android, MacOS, Windows
Retail	Amazon, Alibaba, ...
Transportation & Travel	Uber, Airbnb, TripAdvisor

Source: Parker, Van Alstyne & Choudary

Why Platforms Often Win

- Scale efficiently by removing gatekeepers
 - Example: Amazon's Kindle platform allows anyone to publish & replaces traditional gatekeepers with market signals
- Unlock new sources of value creation & supply
 - Example: Airbnb increases supply of short-term housing
- Use data-driven tools to create community feedback loops
 - YouTube uses user-data to shape recommendations
 - Airbnb uses user feedback for reputation systems

NETWORK EFFECTS

Network Effects: Positive and Negative

- Network effects refer to how the value derived from a good / service / platform depends on the usage of others
- Positive network effects:
 - Friends' participation on social media
 - More uber drivers => more value to riders
- Negative network effects
 - Troll participation on social media
 - More uber riders => more congestion => less value to riders

Two-Sided Network Effects

- Platform often brings together two sides of a market
 - Buyers & sellers, drivers & riders, users & advertisers
- Same-side network effect: How usage of units on same side impact value
 - Often negative in a market settings
 - Sometimes positive: E.g. Facebook users get value from other users
- Cross-side network effect: How usage of units on other side impact value
 - Positive: More drivers is good for riders
 - Negative: Some advertisements are annoying

Demand Economies of Scale

- Many platform businesses have positive network effects
 - Value of platform is increasing in number of users
- Demand economies of scale arise when bigger networks lead to more value created for users
 - Results in an important competitive advantage

Metcalfe's Law

- Value of networks often grow non-linearly with the number of users.
- Example: Telephone network
 - With 1 phone => No connections
 - With 2 phones => 1 connections
 - With 4 phones => 6 connections
 - With 100 phones => 4,950 connections

A Model of Network Effects

- Value of product in market with network effects is given by

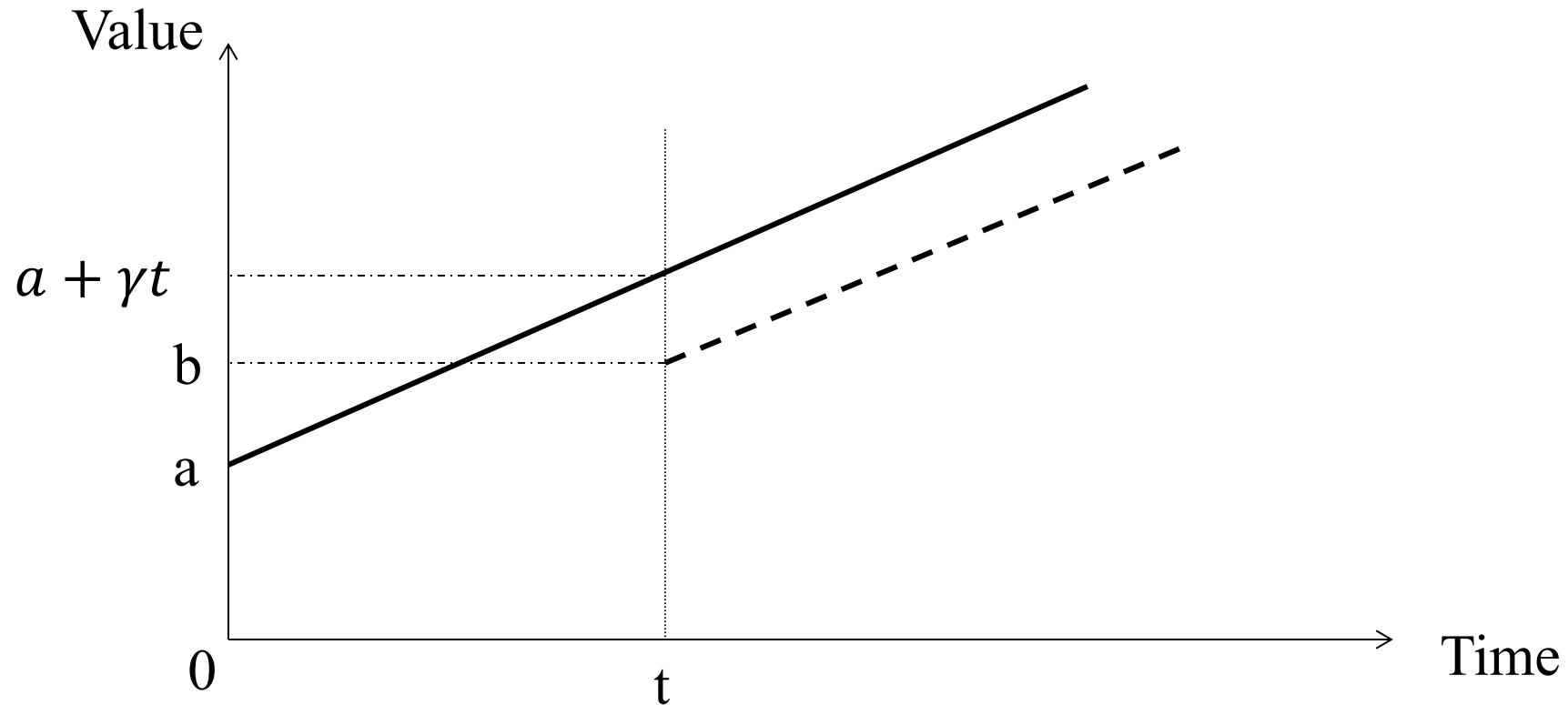
$$V_i = \alpha_i + \gamma Z_t$$

- Z_t : Size of network at time t
- α_i : Value of product to user i without network effects
- γ : Parameter governing how important network effects are

Network Markets: History Matters

- Networks A and B are incompatible but have the same price
- A is available at time 0, B will be available at time t but customers do not anticipate its arrival
- A and B have intrinsic value of a and b respectively, $b > a > 0$
- One customer arrives per period

Network Markets: History Matters, Part 2



Q: Which product will a new customer at time t adopt? Why?

Network Markets: History Matters, Part 3

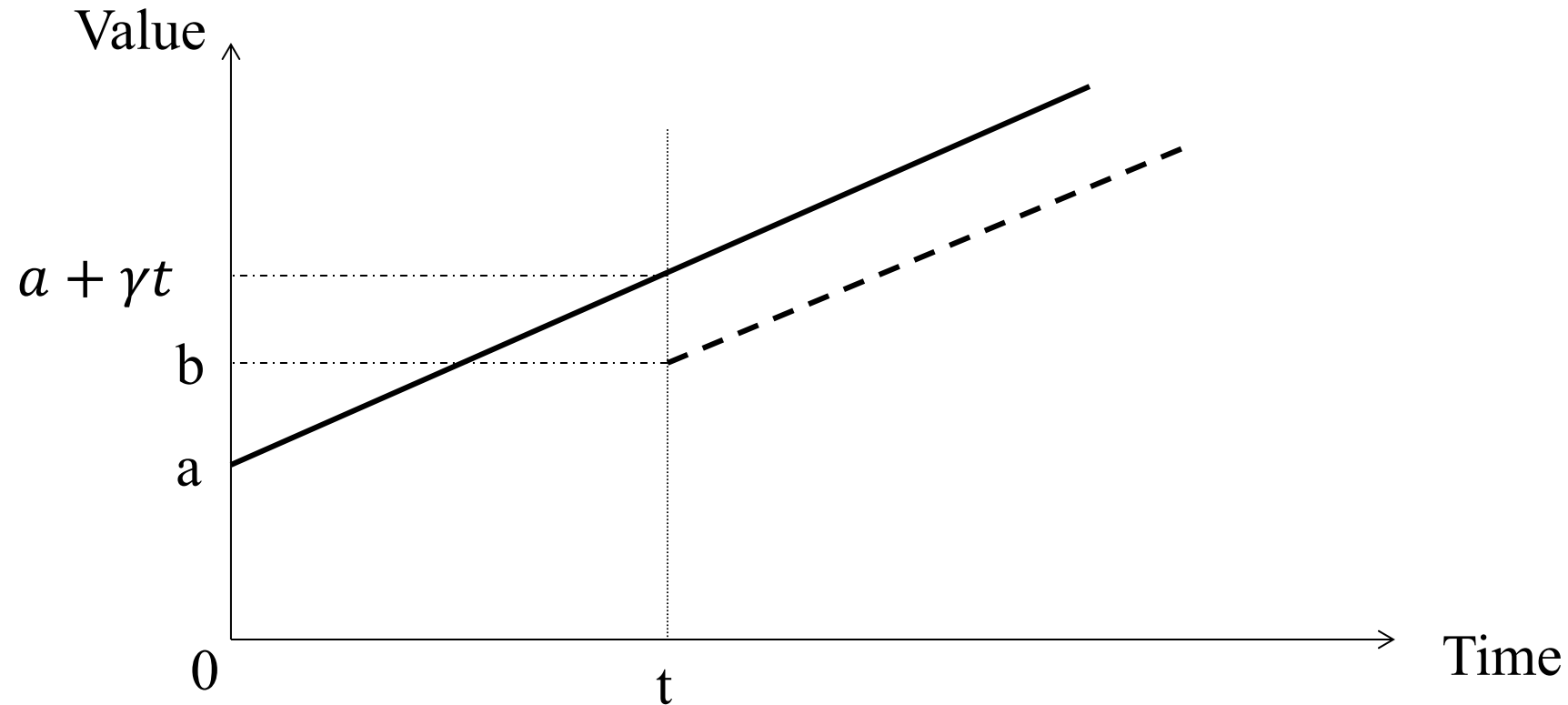
- The superior product B is not adopted
- For network products, both intrinsic performance and installed base matter
- A has inferior product, but has an installed-base advantage by time t

Network Markets: History Matters, Part 4

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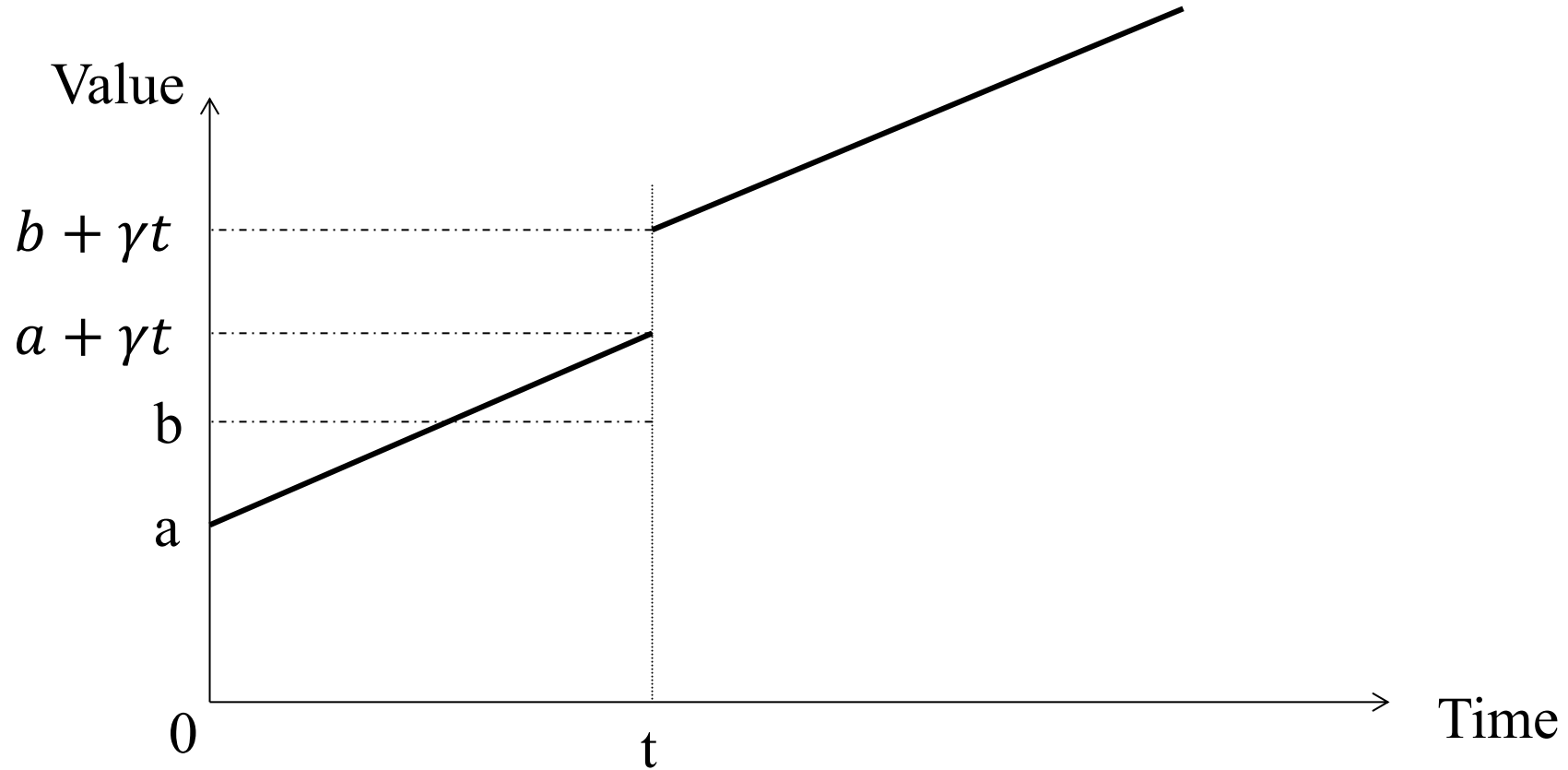
Source: Wikipedia

Network Markets: Beliefs Matter + Multiple Equilibria



Q: How might this graph look different if customers are forward looking?

Network Markets: Compatibility Matters



- Question: What do new customers purchase at time t ? Why?

Tradeoff: Openness vs Control

- A firm benefits from generating network effects if it
 - Is the only supplier of the product (control)
 - Achieves a large user base rapidly (openness)
- Tradeoff:
 - Adoption is more rapid with open standards
 - E.g. Tesla charging standard
 - Profit margins are higher with proprietary standards
 - E.g. Social media, Reddit's third party apps

PLATFORM STRATEGY – LAUNCH & GTM

Launching Platforms

“Prize for greatest salesperson in history should go to whoever sold the first telephone”

- Platforms face a serious chicken & egg problem
 - You need both sides of the market to make platform valuable
 - How to attract one side without the other, and what first?
- Multiple equilibrium abound
 - Just because an idea is good, doesn't mean it will work

Viral Growth

- Traditional products rely on push strategies (e.g. advertising)
- Pull strategies most effective for platforms
 - Need many participants to join and be active (not just sign up) to take advantage of network effects

- Innovative new payment platform, all you need is an email + credit card
- Advertising to potential sellers
 - Too expensive & no demand
- Business development (collaborate with banks)
 - Bureaucratic nightmare
- Solution: Organic viral growth
 - Give people \$10 for signing up and \$10 for referring others

Start as a Non-Platform Business



Single-Side Strategy

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 Location, Restaurant, or Cuisine

Let's go

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ebay Customer Service

Home > ... > Buying > Payments > Paying with PayPal

2 min article

Paying with PayPal

14 May 2008

Android Developer Challenge Judges and Top 50 Details



It's been a busy few weeks here as we've wrapped up the first round of the Android Developer Challenge. We'd like to share a couple pieces of information with you:

- The [full list of judges](#) is now available. It was fun to work with such a diverse group of judges from different companies all around the world.
- A [slide deck of the Android Developer Challenge prize recipients](#) is also available. The deck includes descriptions and screenshots of the 46 recipients who consented to sharing their information and is a great way to get a feel for the quality of apps submitted.

The prize recipients' entries were just the tip of the iceberg in terms of great applications submitted, and we'd like to thank and congratulate everyone who entered the challenge. We look forward to seeing all of the application in the hands of consumers with Android devices.

Marquee Strategy



Podcast Pay Dirt: Why Top Hosts Like Joe Rogan, Kelce Brothers, Alex Cooper Are Landing Big-Bucks Deals

Nine-figure talent pacts reflect health of growing sector – and underscore heightened battle for mass audiences

By Todd Spangler ▾



Rogan: Vivian Zink/Syfy/NBCU Photo Bank/NBCUniversal; Kelces: Courtesy of New Heights; Cooper: Todd Owyong/NBC/Getty Images

SXSW Helped Launch Twitter and They Never Looked Back

 Comments **1**  Email  Print  Share

Does Twitter owe its success to SXSW? That may be a stretch but Twitter can thank the tech and media savvy throngs at SXSW for tripling their traffic at the 2006 event from 20,000 to 60,000 tweets a day. Eight years later and 288 million users later, that number averages a half billion tweets per day. What has surprised Twitter founder Jack Dorsey the most about how people use his creation? Hear the answer to that and much more on this edition of The Business.

Hundreds Register for New Facebook Website

Facemash creator seeks new reputation with latest online project

By Alan J. Tabak, Crimson Staff Writer

February 9, 2004

When Mark E. Zuckerberg '06 grew impatient with the creation of an official universal Harvard facebook, he decided to take matters into his own hands.

After about a week of coding, Zuckerberg launched thefacebook.com last Wednesday afternoon. The website combines elements of a standard House face book with extensive profile features that allow students to search for others in their courses, social organizations and Houses.

PLATFORM STRATEGY – MONETIZATION

How to Monetize

- Transaction fee:
 - Avoids upfront cost of joining platform, only pay when value created
 - Sometimes hard to prevent transactions moving off platform
- Charging for access / enhanced access:
 - Avoids incentive to take transactions off-platform
 - Risk of increasing “low-quality” contributions + barrier to initial signup
 - Advertising is a form of this (pay for promotion)

Exercise: Who to Monetize & What to Charge

- We are a platform business bringing together users/consumers (C) and advertisers (A)

- Example: Social media, content platforms, search

- Consumer value does not depend on advertiser

$$q_C(p_C, p_A) = 100 - 1.1 * P_C$$

- Advertiser value (and hence demand) depends on user base

$$q_A(p_C, p_A) = \gamma q_C(p_C, p_A) - p_A$$

- Zero marginal costs
- Set $\gamma = 1$

Exercise: Treat Markets Independently

- Consumer value does not depend on advertiser

$$q_C(p_C, p_A) = 100 - 1.1 * P_C$$

- Optimal price for consumers alone solves

$$\tilde{p}_C = \arg \max_{p_C} p_C(100 - 1.1 * p_C)$$

First calculate optimal price for consumers ignoring the advertiser side of the market. Create a function that takes price and returns profit.

Consumer-Only Profits



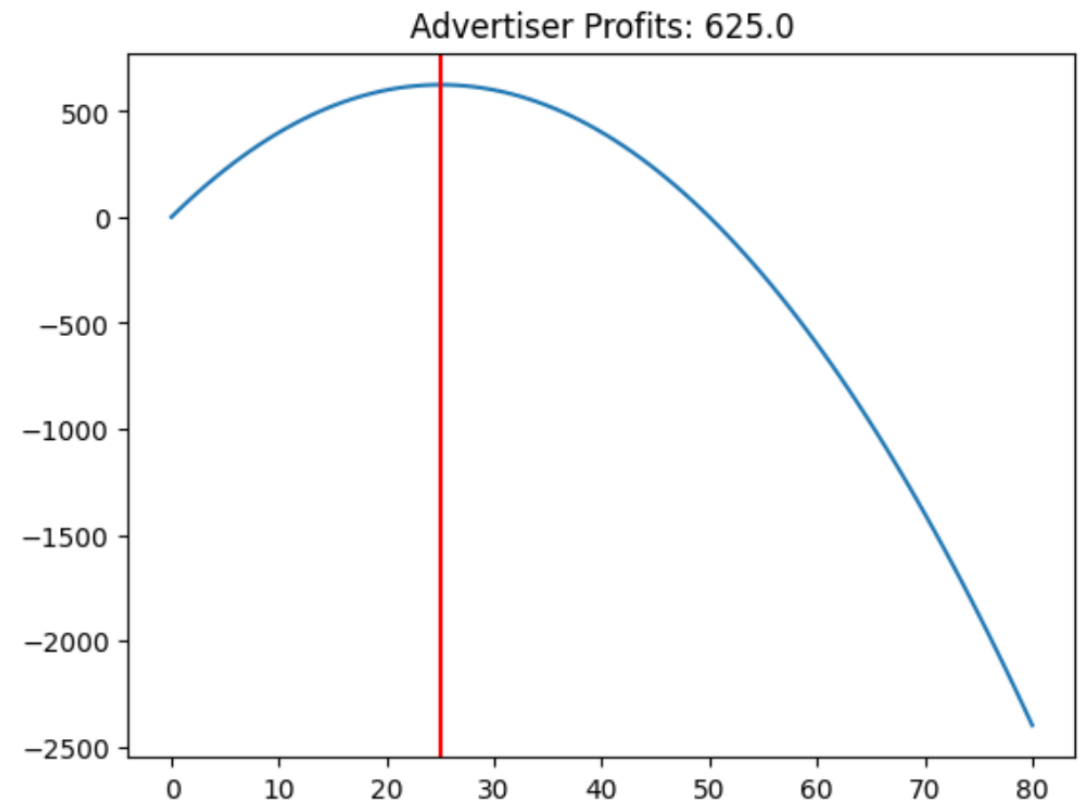
Exercise: Treat Markets Independently, Continued

- Optimal price for consumers alone solves is $\tilde{p}_C \approx 45.5$
- Now let's calculate the optimal price for advertisers, treating the consumer side as fixed

$$\tilde{p}_A = \arg \max_{P_A} P_A(\gamma q_C(\tilde{p}_C) - P_A)$$

1. Calculate $q_C(\tilde{p}_C)$
2. Solve for optimal P_A

Advertiser-Only Profits



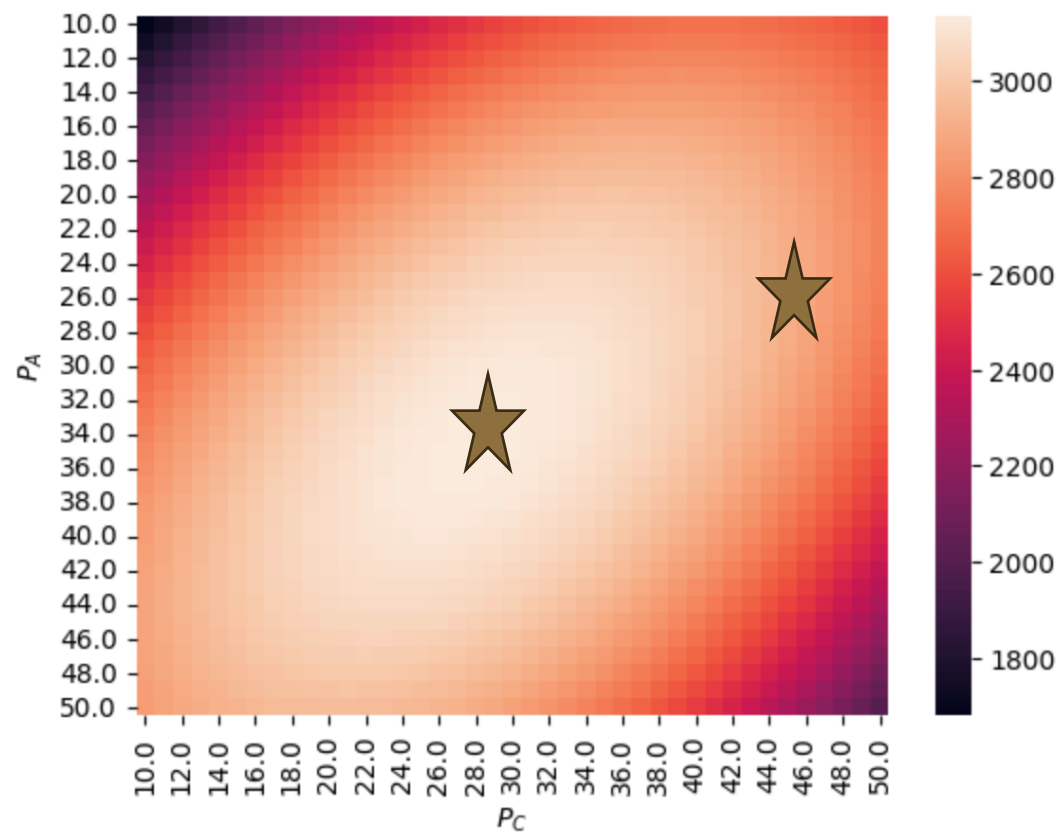
Now Let's Jointly Optimize

- Want to solve for the consumer and advertiser prices that maximize total profits:

$$P_A^*, P_C^* = \arg \max_{P_A, P_C} P_A q_A(P_A, P_C) + P_C q_C(P_C)$$

- Write a function that calculates total profit as a function of prices
 - Helpful to break this down into sub-functions that calculate quantity for each side of the market
- Try different values to see what maximizes profit
 - Optimal prices treating each market as independent are good place to start

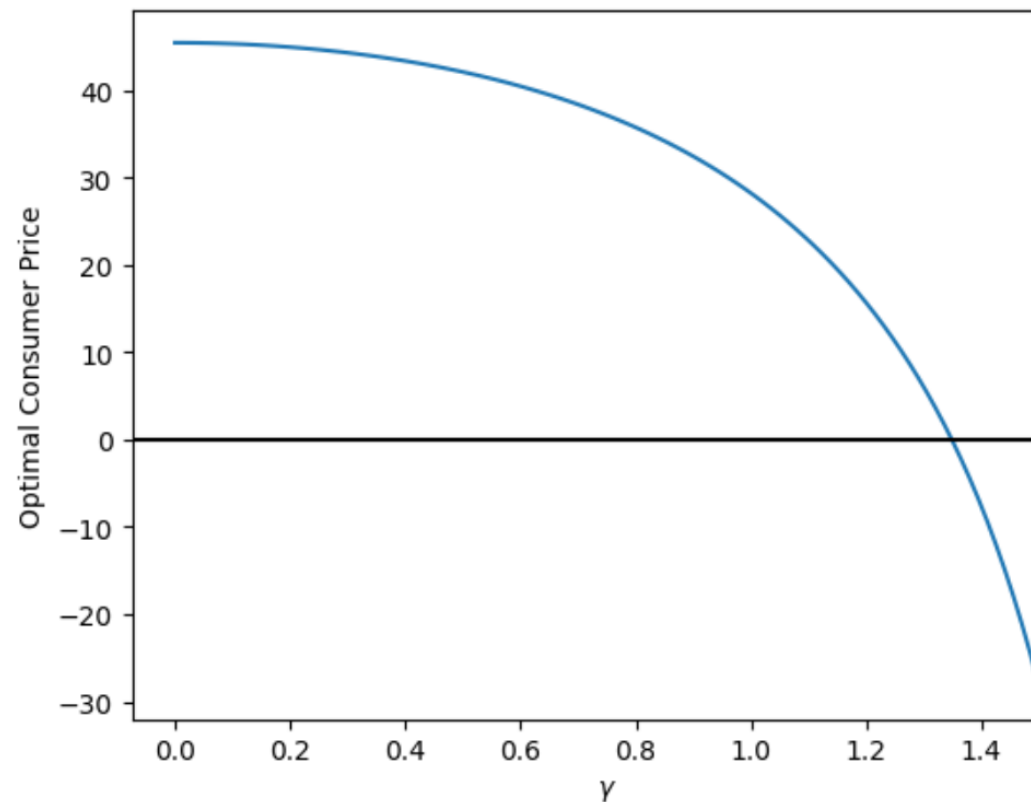
Now Let's Jointly Optimize, Continued



Takeaways

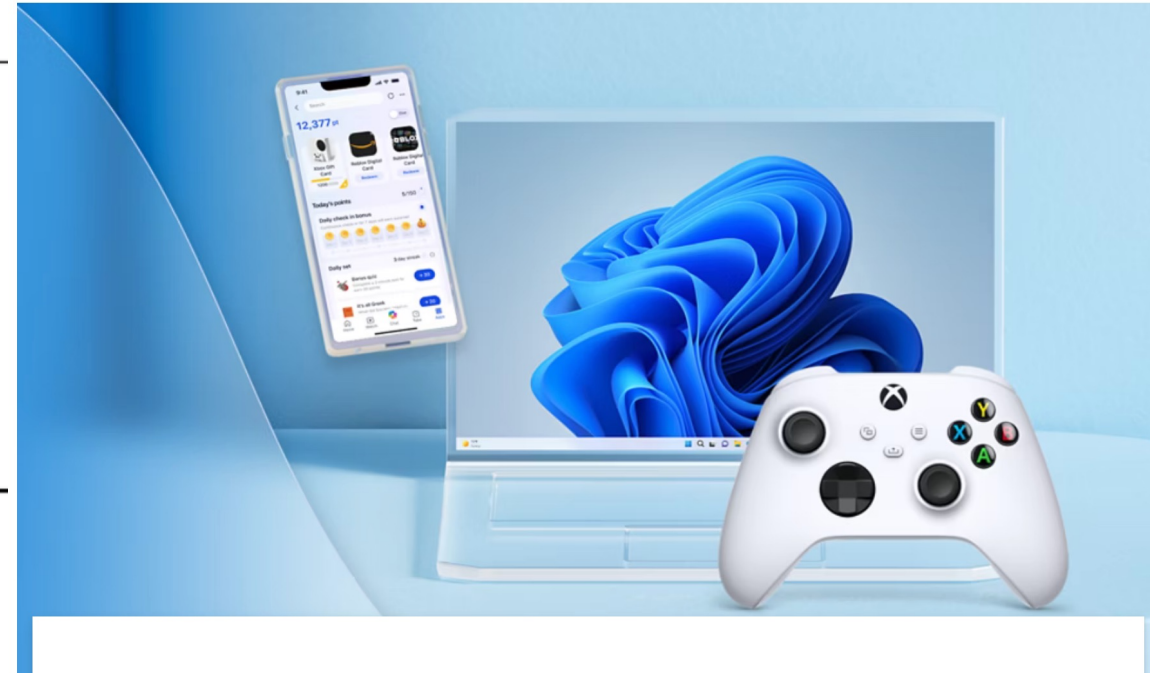
- Network effects from consumers to advertisers
 - Subsidize consumers to encourage participation
 - Raise prices on advertisers to extract value
- More generally:
 - Subsidize side that is (1) price elastic and (2) provides more positive network effects
- Examples:
 - Social media, search, TV
- If network effect large enough, can be optimal to charge 0 or negative price!

What Happens If $\gamma = 1.4$?



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QUESTIONS